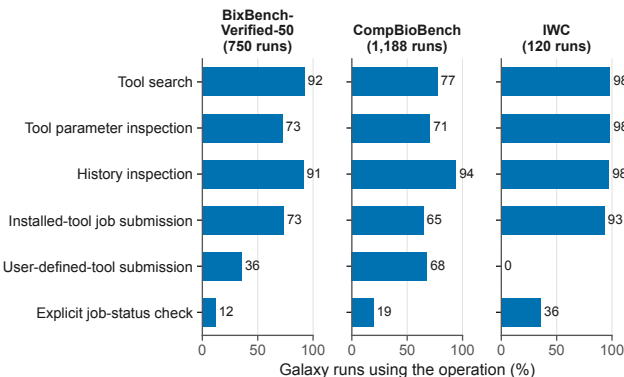
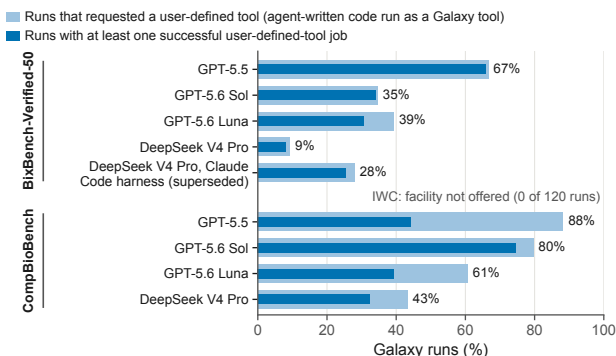


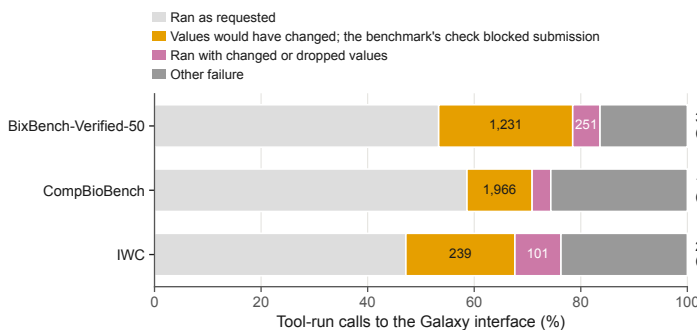
a Agents used every core workflow operation



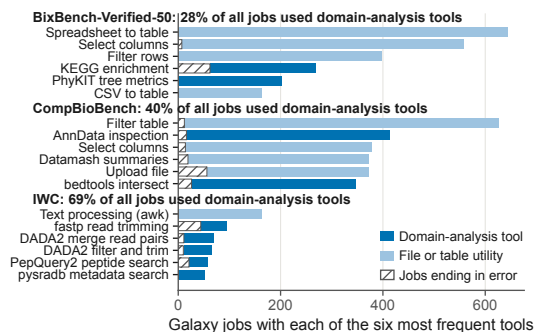
c Use of user-defined tools depends on the model



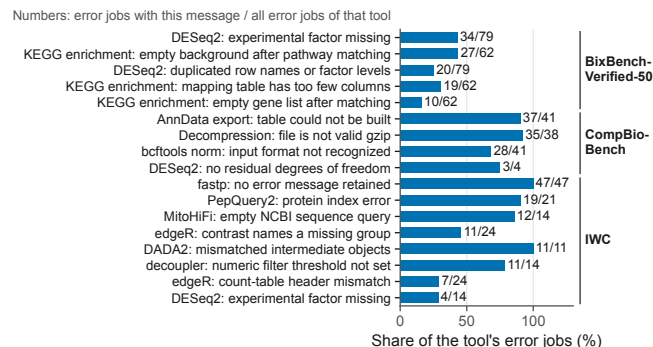
e Galaxy silently changed requested parameter values in 16–30% of tool runs



b Workflow-derived tasks relied more on domain-analysis tools



d Recorded errors point to inputs and parameters, not to missing software



Examples: value requested by the agent and value Galaxy actually used

Tool	Parameter	Requested	Used by Galaxy
Datamash	operation	max	count
Datamash	column number	6	1
PhyKIT metrics	metric	treeness	total tree length
STAR-Fusion	genome	human (hg38)	honey bee (apiMel4)
STAR-Fusion	read layout	paired-end	single-end
Unzip	files to extract	matching pattern	all files
DESeq2	factor level	DMSO	FactorLevel